

SAAD IKRAM

Junior Flutter Developer

+92-314-5459961 | saadnaz43165@gmail.com | Chakwal, Punjab, Pakistan
github.com/saad43165 | portfolio-saad-mu.vercel.app | linkedin.com/in/saad-i-786123406

Available for immediate relocation to Lahore, Islamabad & Rawalpindi

PROFESSIONAL SUMMARY

Junior Flutter Developer with 3 internships and hands-on experience building cross-platform mobile applications using Flutter, Firebase, Supabase, REST APIs, and FastAPI. Also integrated AI-powered features including TFLite models, OCR, and Groq API into production-ready mobile apps. Comfortable with React Native as a secondary ecosystem. Familiar with SQA practices including manual testing, black-box and white-box testing, mutation testing, and test case design. Eager to contribute to a professional team and keep growing as a mobile developer.

Flutter Stack: Flutter, Dart, Firebase, Supabase, REST APIs, Provider, Riverpod, SQLite, Git, Android Deployment

WORK EXPERIENCE

Software Engineering Intern

Saf's Solutions | Rawalpindi, Pakistan

Jun 2025 – Sep 2025

- Developed responsive cross-platform applications in React Native, ensuring seamless UX and functional consistency across Android and iOS.
- Collaborated in agile development cycles, contributing to component architecture, state management, and API integration.
- Participated in code reviews and sprint planning, improving delivery quality and team coordination.

Flutter Development Intern

Saf's Solutions | Rawalpindi, Pakistan

Jun 2024 – Aug 2024

- Built Flutter frontend components and integrated third-party APIs to ensure application effectiveness and reliability.
- Gained hands-on experience with state management patterns and Firebase backend services.
- Assisted in diagnosing and fixing UI/UX issues, reducing reported bugs by iterating on user feedback.

Corporate Trainee Intern

Wah Noble Group of Companies | Wah, Pakistan

Jul 2023 – Sep 2023

- Developed professional skills in a corporate environment through diverse technical and collaborative tasks within structured development cycles.
- Strengthened teamwork and communication skills by contributing to cross-functional team deliverables.

PROJECTS

AgriGuard: AI-Powered Plant Disease Detection (FYP) Aug 2025 – May 2026

Flutter · TensorFlow Lite · EfficientNetB3 · Firebase · Python · Dart

- Built a Flutter application with Firebase integration, offline-first architecture, cloud synchronisation, and real-time image processing — deployed to low-end Android devices for remote farmers.
- Engineered a hybrid deep learning framework using EfficientNetB3 and TFLite, achieving 94.8% classification accuracy within a 3.1 MB model optimised for low-end Android devices.
- Implemented a 5-part diagnostic system: severity quantification via pixel-ratio analysis, localised remediation guidance, and multimodal API for out-of-distribution (OOD) species detection.
- Integrated Firebase for cloud-based Advanced AI scan mode, enabling seamless online/offline fallback for farmers with intermittent connectivity.

AdverShield: AI-Powered Android Malware Detection Jun 2025 – Aug 2025

Flutter · Dart · Python · FastAPI · TensorFlow Lite · SQLite

- Built a Flutter mobile security application with SQLite persistence, real-time threat dashboards, and an interactive analytics UI — scanning installed Android apps for malware, spyware, and adware.
- Engineered a hybrid AI inference pipeline combining an edge-optimised TensorFlow Lite model and a cloud Python backend, analysing 215 distinct DREBIN dataset features for high-accuracy threat classification.

- Developed a high-performance RESTful API using FastAPI to securely process complex feature scaling and predictive analytics, seamlessly integrating with the mobile client.
- Built a dynamic, data-driven Flutter UI featuring real-time security scoring, interactive analytics dashboards, and local scan history persistence via SQLite.

SignBridge: Two-Way AI Sign Language Translation 2025

Breaking the Silence — Two-Way AI Sign Language Translation

Flutter · Dart · Google ML Kit · TensorFlow Lite · Three.js · Drift (SQLite) · TTS/STT

- Engineered a dual-direction translation app bridging the deaf-mute community and the hearing public — Receptive Mode (Sign-to-Text via live camera) and Expressive Mode (Text/Voice-to-Sign via rigged 3D avatar).
- Implemented real-time hand gesture classification using Google ML Kit Pose Detection and a custom on-device TFLite neural network with adaptive smoothing algorithms to eliminate frame jitter.
- Integrated an interactive 3D avatar via WebGL-accelerated WebViews (Three.js / model-viewer), with a single-instance ValueKey listener to eliminate WebView memory bloat on model swap.
- Engineered a Greedy Text Tokenizer with a stop-words filter and regex phrase matcher — converts conversational sentences to root signs, preventing character-by-character fingerspelling.

MediVision: AI-Powered Healthcare Assistant Mar 2025 – May 2025

Flutter · FastAPI · Python · SQLite · Capacitor · OCR · FDA Drug API

- Built a cross-platform Flutter health tracking app with secure JWT authentication, role-based access control, FastAPI backend, and a responsive premium UI across Android and iOS.
- Integrated an AI Prescription Scanner using OCR to digitise handwritten medical data and queried the FDA Drug Database to alert users of dangerous drug interactions in real time.
- Developed a predictive linear regression engine in Python to analyse historical patient vitals and forecast 30-day health trajectories, including hypertension and chronic risk scoring.
- Designed a modular FastAPI backend with SQLite persistence, exposing RESTful endpoints consumed by both the mobile app and a Capacitor-powered web interface.

VaxGuard: AI-Powered Clinical Decision Support System Jan 2025 – Mar 2025

Flutter · Dart · Provider · Local Storage · REST APIs

- Built a cross-platform Flutter app with Provider state management, local persistence, REST API integration, and multi-language support (English / Urdu) for diagnostic triage and vaccination management.
- Integrated a WHO 2018 Rabies Protocol-compliant Animal Bite Assessor evaluating bite location and exposure severity, auto-generating personalised Post-Exposure Prophylaxis (PEP) schedules.
- Built a real-time geospatial Live Outbreak Radar to monitor active disease outbreaks near the user, with push notifications for proximity alerts.
- Implemented a secure local persistence layer to maintain full patient history, diagnostic assessments, and vaccination records across sessions.

RunQuest: Gamified Territory Conquest & Fitness App Oct 2024 – Dec 2024

React Native · Expo · TypeScript · Firebase · Zustand · Turf.js

- Architected a gamified fitness app where users track real-world runs to claim virtual territories on a live map, combining geolocation, health metrics, and multiplayer competition.
- Engineered territory conquest mechanics using Turf.js and React Native Maps — processing live GPS data to calculate closed-loop paths, polygon areas, and competitive takeovers in real time.
- Built a live multiplayer ecosystem with Firebase: real-time presence tracking, global team-based leaderboards, and a dynamic activity feed for sustained user engagement.
- Designed a premium UI with Zustand state management, Expo Reanimated micro-interactions, real-time weather, dynamic map themes, achievement system, and a custom in-app music player.

Naheed & Sons: Full-Stack Corporate Web Application & PWA Jul 2024 – Sep 2024

naheedandsons.com.pk

Next.js 16 · React 19 · TypeScript · Tailwind CSS · Framer Motion · Supabase · PWA

- Architected a high-performance, SEO-optimised corporate web application and PWA for a premium construction and engineering firm using the Next.js App Router.
- Engineered a secure, password-protected Admin CMS utilising Supabase for persistent cloud storage and intelligent localStorage caching, enabling zero-latency content management and offline resilience.
- Developed an interactive, multi-step Budget Calculator and Quote Generation system that automates lead qualification and securely processes client inquiries in real time.

- Implemented advanced PWA architecture featuring custom service workers and native installation prompts, delivering a seamless, app-like mobile experience alongside complex Framer Motion UI animations.

GitHub Repository Explorer Apr 2024 – May 2024

Flutter · Dart · GitHub REST API · SharedPreferences

- Built a mobile app interfacing with the GitHub REST API to search and explore repositories by name, language, or owner with infinite-scroll pagination.
- Built a local storage ecosystem using SharedPreferences — persisting search history and a favourites system for offline tracking of bookmarked repositories.
- Designed a glassmorphism-inspired UI with micro-animations, gradient themes, dark/light mode toggle, and bottomsheet modals for intuitive navigation.

EDUCATION

Bachelor of Software Engineering

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology, Mang, Haripur Oct 2022 – May 2026

HSSC – Pre-Engineering (Grade: A1)

Mulhal Science College 2019 – 2021

SSC – Matriculation (Grade: A1)

Blooming Scholars Academy, Mulhal 2017 – 2019

TECHNICAL SKILLS

Mobile Development: Flutter (Dart) · Firebase · Supabase · REST API Integration · SQLite · Firestore

State Management: Provider, Riverpod

AI / ML: TensorFlow Lite, EfficientNetB3, OCR, AI Model Integration

Backend: FastAPI (Python), RESTful API Design

Dev Tools: Git / GitHub, Android Studio, Android Build & Deployment

Cross-Platform: React Native (Expo, TypeScript)

Web Development: Next.js, React, TypeScript, Tailwind CSS, PWA, Framer Motion

Quality Assurance: Manual Testing, Black-Box & White-Box Testing, Mutation Testing, Test Case Design, Bug Reporting, Playwright

LANGUAGES

Urdu: Native · Punjabi: Native · English: Professional Proficiency

REFERENCES

Available upon request.